

Exercise 1: Introduction

Submission: Sunday, 15th March, 23:59h

Task 0: Get to know Plotly / Dash

0 points

In the exercises, we are going to use Plotly Dash (<https://dash.plotly.com/>) and Python. In the following, we will set up a conda environment for the exercises' coding tasks.

- Download the file **environment.yml** from OLAT.
- If you haven't installed conda already, install it following the instructions on the official website (<https://docs.conda.io/en/latest/>).
- In your terminal, run (from the directory where the **environment.yml** file is stored):

```
conda env create -f environment.yml
```

This will create a conda environment called '**dvc**'. It contains all the necessary packages to complete the coding exercises.

- To **activate** the environment, run:

```
conda activate dvc
```

This has to be repeated every time you reopen your terminal. You should have the environment activated every time while executing the coding exercises for this lecture. This ensures that we are able to reproduce the results of your solutions.

- Try running some small tests. We recommend that you go through the tutorial provided for Dash (<https://dash.plotly.com/tutorial>). You don't have to do all of it, but it gives a good overview of the main functionalities of Dash.

Task 1: First visualization

In this task, we will look into the basic functionalities of plotly dash by creating a line chart of a given dataset.

2 points

- Download** the dataset given at <https://www.kaggle.com/datasets/patrlris/crop-yield->

prediction-dataset. We will use the file **yield.csv** in this exercise. In the dataset, the relevant columns are:

Area: country name

Item: crop name

Year: year

Value: yield in hg/ha

- b. Read in the dataset using **pandas** (<https://pandas.pydata.org/>).
- c. Use Dash to create a **line chart** to display all crops yield over time with clear legend to show the information of country, year period, and crops as well as hover-over information of each data point. The year should be on the x-axis and the value on the y-axis.
- d. Implement the plot as a function, which accept the country name and year period as parameters. The selected country and year range should dynamically determine which crops are displayed and their corresponding yield values.

Hand-in

Please hand in your solution in OLAT as a **.zip** file until **Sunday, 15th March, 23:59h**. The .zip file should include the following:

- Your **python code** with the runnable final application in a single file with a reasonable amount of comments explaining your implementation.

Note: You are **not** allowed to use **other libraries** except for the ones stated.

- Include at least two **screenshots** demonstrating your running application on two different countries in two different year periods.

Note: Submissions without screenshots will result in a **0.5 point deduction**.

- In case there are additional remarks, include them as a .txt or .pdf file.
- Please hand-in you own unique solutions. We do not accept copied solutions or solutions created by or with ChatGPT!